MILITARY SPECIFICATION SHEET

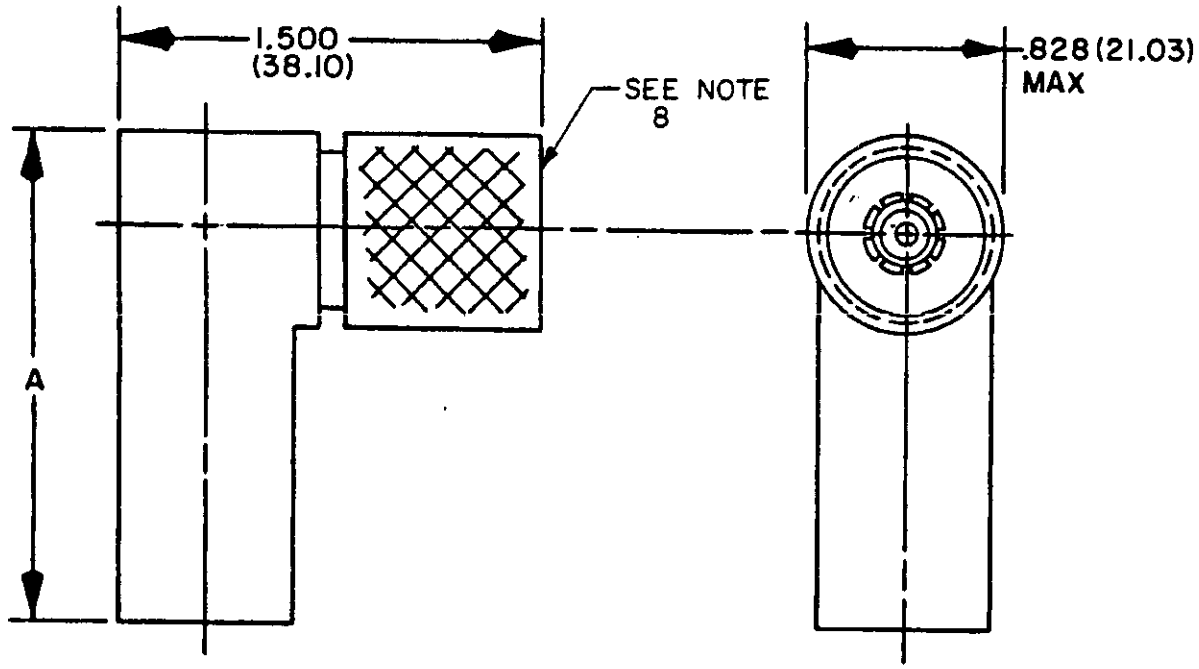
CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY,

(SERIES SC (CABLED), PIN CONTACT, RIGHT ANGLE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-39012.

MARKING IMPLEMENTATION DATE FOR CATEGORY B, SEE TABLE IV



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. For dimensions A, see table I and III.
4. Dimensions A is the maximum envelope dimension.
5. Wrench flats are to accommodate standard wrench opening in accordance with H-28, appendix 10.
6. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
7. All undimensioned pictorial representations are for reference purposes only.
8. Series SC, pin contact interface in accordance with MIL-STD-348, 309.1.

FIGURE 1. General configurations.

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TABLE I. Dash numbers, cross-reference, and dimensions.

Dash number 1/	Applicable cable 2/ M17/	Dimensions	Inches (millimeters) Maximum
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED)			
0101 3/	65-RG165 74-RG213 75-RG214 4/ RG-225/U 5/	A	1.875 (47.63)
0106 3/	74-RG215 6/		
0107 3/	92-RG115 5/		
CATEGORY C - FIELD REPLACEABLE (MIL-C-22520 CRIMP TOOL) 7/ SEE FOOTNOTE NEXT TO APPLICABLE CABLE FOR CRIMP DIE			
0004 3/	65-RG165 5/ 8/ 74-RG213 4/ 8/	A	2.125 (53.98)
0005 3/	75-RG214 4/ 8/ RG-225/U 5/ 8/		
0009 3/	92-RG115 5/ 8/		
CATEGORY D - FIELD REPLACEABLE - DEFINED PIECE PARTS 7/ 9/ 10/			
0501 3/	75-RG214 4/ RG-225/U 5/ 127-RG393	A	2.125 (53.98)
0502 3/	74-RG213 4/		
0503 3/	60-RG142 4/ 5/ 128-RG400		

1/ For cross-reference of dash number to superseded part number or type designation, see table IV.

2/ The latest version of each cable shall be applicable.

3/ These connectors have captivated center contacts.

4/ Cable to be used when performing tests requiring cable except as in 5/.

5/ Cable to be used for the +200°C temperature cycling tests.

6/ Armored cable.

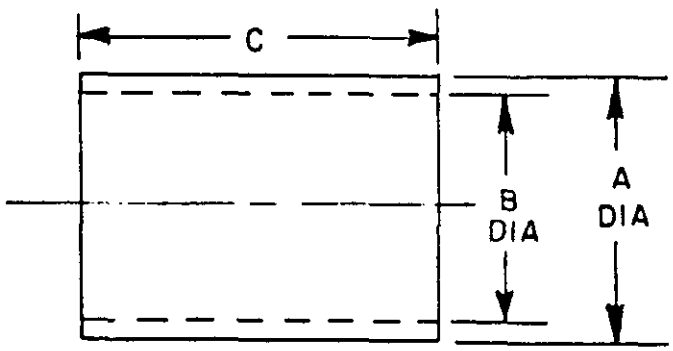
7/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

8/ M22520/5-61 is preferred die.

9/ Not to be used in Army equipment.

10/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

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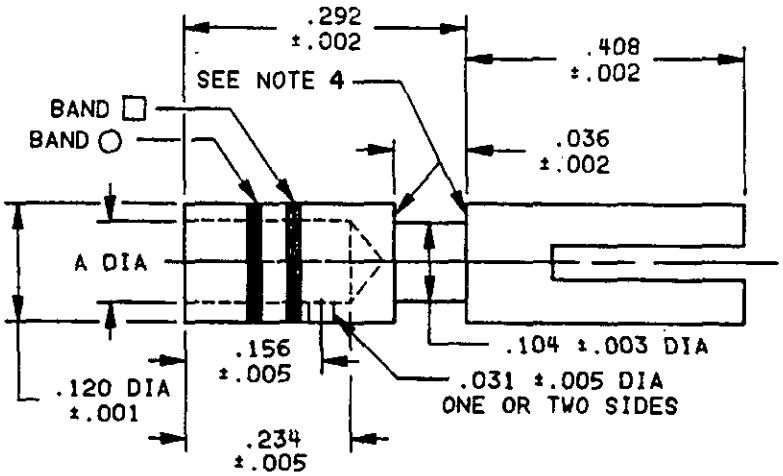


Dash no.	Ferrule no. 2/	A	B	C ±.015	Basic crimp tool 1/ M22520/5-	Crimp die or positioner M22520/5-
0501	39-50	.492 +.003 -.000	.438 +.003 -.000	.600	01	25 Closure A or 61
0502	39-51	.492 +.003 -.000	.418 +.003 -.000	.600		
0503	39-52	.250 ±.003	.220 ±.003	.500		05, 11, 57 Closure A or 19 Closure B

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

FIGURE 2. Contact and ferrule dimensions for category D only.

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CENTER CONTACT

Inches	mm	Inches	mm
.001	0.03	.234	5.94
.002	0.05	.250	6.35
.003	0.08	.292	7.42
.005	0.13	.351	8.92
.015	0.38	.375	9.52
.031	0.79	.394	10.01
.036	0.91	.406	10.31
.043	1.09	.408	10.36
.098	2.49	.418	10.62
.104	2.64	.438	11.13
.120	3.05	.492	12.50
.156	3.96	.500	12.70
.220	5.59	.600	15.24

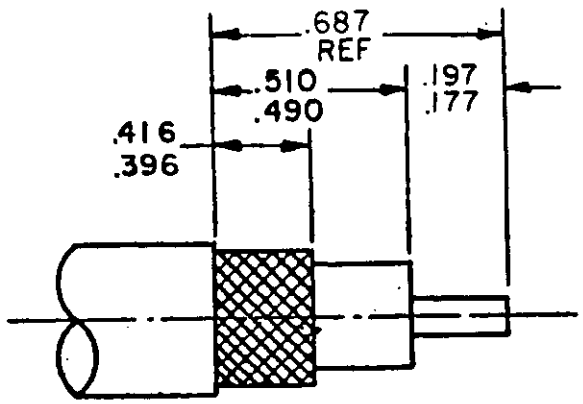
Dash no.	Contact no. 2/	A	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
0501 0502	39-10	.098 ±.002	M22520/1-01	M22520/1-13	60 lbs	Red	Orange
0503	39-11	.043 +.001 -.002	M22520/1-01	M22520/1-13	20 lbs	Blue	

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Crimp tensile test shall be in accordance with MIL-C-39029.
4. .003 maximum break.
5. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
6. Form socket to mate with .063/.066 diameter pin.

FIGURE 2. Contact and ferrule dimensions for category D only - Continued.

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Inches	mm
.177	4.50
.197	5.00
.396	10.06
.416	10.57
.490	12.45
.510	12.95
.687	17.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

1,000 volts rms maximum working voltage at sea level.

250 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Design and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force - Not applicable.

Torque - 6 inch-pounds maximum.

Coupling proof torque: 15 inch-pounds.

Inspection conditions: Coupling torque - 6 - 10 inch pounds.

Mating characteristics: See MIL-STD-348 for dimensions.

Outer contact:

Test ring ID - .411 maximum, 16 microinch finish.

Insertion force - 15 pounds maximum when inserted a minimum of .125.

Contacts with slotted members: Shall contact a .419 minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms minimum.

Center contact retention: (Applicable to captivated-center-contact connectors only.) 15 pounds minimum axial force for all cable except RG400 and RG142; 6 pounds minimum for RG400 and RG142.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR):

From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower.

From 500 to 9,000 MHz: 1.35 maximum.

From 9,000 to 11,000 MHz: 1.50 maximum.

Swept frequency VSWR test setup:

Item 6 - VSWR shall be less than 1.015 +.005 F (F in GHz).

Item 16 - VSWR shall be less than 1.015 +.005 F (F in GHz).

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Second step of VSWR checkout procedure - VSWR shall be less than 1.045 +.019 F (F in GHz).

Group B inspection - VSWR shall be less than 1.10 +.01 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.15.

Connector durability:

500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact	2.0	2.5
Outer contact	.15	Not applicable
Braid to body	.05	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 3,000 volts rms at sea level for connectors using other than RG400 and RG142; 1,500 volts rms minimum for connectors using RG400 and RG142.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage - 750 volts rms minimum.

Altitude - 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency - 2,500 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current - Not applicable.

Cable retention force:

Noncrimp assemblies: 75 pounds minimum.

Crimp assemblies:

50 lbs minimum for cables .155-.189 OD.

60 lbs minimum for cables .190-.229 OD.

75 lbs minimum for cables .230-.249 OD.

90 lbs minimum for cables .250 OD and larger.

Coupling mechanism retention force: 100 pounds minimum.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz. This requirement may be met by qualifying the straignt connector version for the same cable under MIL-C-39012/35.

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Insertion loss:

.21 dB maximum at 9 GHz.

.07 $\sqrt{F\ (GHz)}$ dB maximum tested at 3 GHz and 6 GHz.

Part number: M39012/39- (dash number from table I or "B" number from table III).

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/39	M39012/39
I	-0101	-0101 -0106 -0107
II	B0002	B0002 B0003 B0008
III	-0004	-0004 -0005 -0009
IV	-0501 -0502	-0501 -0502 -0503
V	-0503	-0503

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

TABLE III. CATEGORY B - NONFIELD REPLACEABLE (SPECIAL TOOLS MAY BE REQUIRED). 1/

| NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY. |

Part number M39012/39B 2/3/	Applicable cable 4/ M17/	Dimensions	Inches (millimeters) Maximum
0002	65-RG165 5/ 74-RG213 6/	A	2.125 (53.98)
0003	75-RG214 5/ RG-225/U 5/		
0006	92-RG115 5/ 112-RG304		

1/ For maintenance replacements for category B, see table V.
2/ For cross-reference of dash number to superseded part number or type designation, see table IV.
3/ These connectors have captivated center contacts.
4/ The latest version of each cable shall be applicable.
5/ Cable to be used for the +200°C temperature cycling tests.
6/ Cable to be used when performing tests requiring cable except as in 5/.

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TABLE IV. Cross-reference of part numbers. 1/

Preferred part number	Substitute for part number or type designation 2/
M39012/39	M39012/39
-0101	-0001
-0106	-0006
-0107	-0007
B0002	-0002
B0003	-0003
B0008	-0008

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The connectors that are in stock or distribution that were previously qualified and marked with the old part number shall also be considered acceptable for Government use until connector stock is purged.

2/ The superseded part number or the type designation is for cross-reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/39-XXXX shall be used in all cases for marking and identifying the connector.

TABLE V. Maintenance replacements for category B.

Category B number 1/	Category C dash number	Category A dash number	Category D dash number
B0002	0004	0101	0502
B0003	0005	0101	0501
B0008	0009	0107	---

1/ Category B connectors are for original installation only. They will not be stocked or procured by the Government.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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